System Validation Report

Project	Online Exam Portal (Software Engineering mini project)
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Build	`main` at commit `2f93f79`
Test run	11-11-2026 21:00 IST
Generated by	`docs/reports/generate_reports.py` (do not edit by hand; run the script again)
References	SRS v1.0, SAD v1.0, Software Test Plan v1.0, Test Case Document

1. Purpose

This report records whether the Online Exam Portal meets the requirements in the SRS, by executing every test case in the Software Test Plan (Section 13 RTM) against the build above. Each test case is validated by automated tests, by a recorded manual or tool-based run, or both. The detailed steps and expected results of each test case are in the Test Case Document.

2. Test environment

Item	Value
OS	Linux 6.18.44-fc-v378
Python	3.13.18
Django / DRF	5.2.18 / 3.18.3
Test runner	pytest 9.1.1 with pytest-django, pytest-cov
Coverage tool	coverage.py 7.17.2 (line + branch, settings in `.coveragerc`)
Database	SQLite (Django test database, created fresh for the run)
Command	`python docs/reports/generate_reports.py` (wraps `pytest --cov --junitxml`)

Browser checks (`docs/browser_checks/`) run the real website in a headless browser with axe-core. The availability run uses `deploy/uptime_probe.py` against a running server.

3. Validation results by test case

Test case	Requirement	Title	Priority	Method	Automated passed	Evidence	Status
TC-Auth-01	OEP-F-001	Student registers with a valid email or roll number	High	Automated	17/17	Automated tests (see Unit Test Report)	**Passed**
TC-Auth-02	OEP-F-002	User logs in with username, email or roll number	High	Automated	24/24	Automated tests (see Unit Test Report)	**Passed**
TC-Auth-03	OEP-F-003	Each role reaches only its own screens and APIs	High	Automated	9/9	Automated tests (see Unit Test Report)	**Passed**
TC-Auth-04	OEP-F-004	Five consecutive failed logins lock the account	Medium	Automated	12/12	Automated tests (see Unit Test Report)	**Passed**
TC-QB-01	OEP-F-010	Faculty creates questions of each type	High	Automated	18/18	Automated tests (see Unit Test Report)	**Passed**
TC-QB-02	OEP-F-011	Questions are tagged and can be filtered	Medium	Automated	8/8	Automated tests (see Unit Test Report)	**Passed**
TC-QB-03	OEP-F-012	Faculty imports questions from the template	Medium	Automated	12/12	Automated tests (see Unit Test Report)	**Passed**
TC-QB-04	OEP-F-013	Edit lock on questions in use	Medium	Automated	10/10	Automated tests (see Unit Test Report)	**Passed**
TC-Exam-01	OEP-F-020	Faculty assembles an exam manually and by rules	High	Automated	19/19	Automated tests (see Unit Test Report)	**Passed**
TC-Exam-02	OEP-F-021	Exam scoring settings are saved and validated	High	Automated	8/8	Automated tests (see Unit Test Report)	**Passed**
TC-Exam-03	OEP-F-022	Exam opens and closes on its window	High	Automated	4/4	Automated tests (see Unit Test Report)	**Passed**
TC-Exam-04	OEP-F-023	Each student gets an independently shuffled paper	Medium	Automated	11/11	Automated tests (see Unit Test Report)	**Passed**
TC-Exam-05	OEP-F-024	Faculty assigns an exam to a course and publishes it	High	—	—	—	**Not executed**
TC-Deliver-01	OEP-F-030	Exam screen with server-synced timer	High	Automated	22/22	Automated tests (see Unit Test Report)	**Passed**

Test case	Requirement	Title	Priority	Method	Automated passed	Evidence	Status
TC-Deliver-02	OEP-F-031	Navigator and flag for review	Medium	Automated + manual	5/5	browser_checks/results.md	**Passed**
TC-Deliver-03	OEP-F-032	Attempt is submitted automatically on timeout	High	Automated	2/2	Automated tests (see Unit Test Report)	**Passed**
TC-Deliver-04	OEP-F-033	Student submits early	High	Automated	2/2	Automated tests (see Unit Test Report)	**Passed**
TC-Deliver-05	OEP-F-034	Single-attempt rule and re-attempt allowance	High	Automated	7/7	Automated tests (see Unit Test Report)	**Passed**
TC-Deliver-06	OEP-F-035	Answers are auto-saved and survive a disconnect	Medium	Automated + manual	3/3	browser_checks/results.md	**Passed**
TC-Eval-01	OEP-F-040	Objective answers are graded on submission	High	Automated	16/16	Automated tests (see Unit Test Report)	**Passed**
TC-Eval-02	OEP-F-041	Pending written answers appear in the grading queue	High	Automated	13/13	Automated tests (see Unit Test Report)	**Passed**
TC-Eval-03	OEP-F-042	Faculty grades a written answer	Medium	Automated	21/21	Automated tests (see Unit Test Report)	**Passed**
TC-Eval-04	OEP-F-043	Final attempt score calculation	High	Automated	12/12	Automated tests (see Unit Test Report)	**Passed**
TC-Result-01	OEP-F-050	Student sees their scorecard	High	Automated	14/14	Automated tests (see Unit Test Report)	**Passed**
TC-Result-02	OEP-F-051	Faculty/Admin view the class report	Medium	Automated	13/13	Automated tests (see Unit Test Report)	**Passed**
TC-Result-03	OEP-F-052	Faculty export the report	Medium	Automated	18/18	Automated tests (see Unit Test Report)	**Passed**
TC-Result-04	OEP-F-053	Faculty view percentage correct per question	Low	Automated	10/10	Automated tests (see Unit Test Report)	**Passed**
TC-Admin-01	OEP-F-060	Admin manages Faculty and Student accounts	High	Automated	26/26	Automated tests (see Unit Test Report)	**Passed**
TC-Admin-02	OEP-F-061	Admin creates courses and maps Faculty/Students	Medium	Automated	20/20	Automated tests (see Unit Test Report)	**Passed**
TC-Admin-03	OEP-F-062	Admin retrieves the audit log	Medium	Automated	6/6	Automated tests (see Unit Test Report)	**Passed**
TC-Admin-04	OEP-F-063	Admin checks system health	Low	Automated	14/14	Automated tests (see Unit Test Report)	**Passed**

Test case	Requirement	Title	Priority	Method	Automated passed	Evidence	Status
TC-Perf-01	OEP-NF-001	200 concurrent exam starts, P90 ≤ 2 s	High	Automated + manual	0/1	performance_report.md	**Not executed**
TC-Rel-01	OEP-NF-002	Saved responses survive client loss	High	Automated + manual	1/1	browser_checks/results.md (offline/reconnect, reload/resume)	**Passed**
TC-Ops-01	OEP-NF-003	Uptime during a simulated exam window	High	Automated + manual	15/15	availability_report.md	**Passed**
TC-Audit-01	OEP-NF-004	Audit logging and retention	Medium	Automated	23/23	Automated tests (see Unit Test Report)	**Passed**
TC-UX-01	OEP-NF-005	Usability and accessibility of the exam website	Medium	Automated + manual	14/14	browser_checks/results.md, usability_test.md	**Partly passed**
TC-Arch-01	OEP-NF-006	Architecture review against the SAD	Medium	Manual	—	architecture_review.md	**Partly passed**
TC-Sec-01	OEP-SR-001	TLS enforcement	High	Automated + manual	17/17	tls_deployment.md (check_tls.sh run on 11-11-2026: 11/11 checks passed)	**Passed**
TC-Sec-02	OEP-SR-002	Password hashing	High	Automated	6/6	Automated tests (see Unit Test Report)	**Passed**
TC-Sec-03	OEP-SR-003	Session expiry and invalidation	High	Automated	14/14	Automated tests (see Unit Test Report)	**Passed**
TC-Sec-04	OEP-SR-004	Answer-key confidentiality	High	Automated	15/15	Automated tests (see Unit Test Report)	**Passed**
TC-Sec-05	OEP-SR-005	Lockout audit trail	Medium	Automated	4/4	Automated tests (see Unit Test Report)	**Passed**
TC-Sec-06	OEP-SR-006	Server-side role check on every endpoint	High	Automated + manual	17/17	endpoint_roles.md	**Passed**

4. Summary

Metric	Value
Test cases planned	43
Executed	41 (95.3%)
Not executed	2
Partly passed	2
Passed	39

Requirement coverage

Category	Requirements	Fully validated	Partly validated	Not validated
Functional (OEP-F)	31	30	0	1
Non-functional (OEP-NF)	6	3	2	1
Security (OEP-SR)	6	6	0	0

5. End-to-end exam lifecycle

The Test Plan's system test is the full lifecycle: registration → question bank → exam creation → delivery → grading → results. It is covered by:

- `evaluation/test_scorecard.py::EndToEndTests`: a student sits a mixed exam, faculty grade the written answer, the student opens the scorecard.
- `docs/browser_checks/` "Full sitting" check: in a real browser a student answers, flags, navigates and submits; a second start is blocked.
- The per-step test cases in Section 3 (TC-Auth-01, TC-QB-01, TC-Exam-01, TC-Deliver-01…06, TC-Eval-01…04, TC-Result-01…04).

Result: `test_full_flow` **passed**.

6. Security validation (Test Plan §5.1)

§5.1 check	Test case	Automated passed	Evidence	Status
Password hashing with bcrypt/Argon2, never logged or plaintext (OEP-SR-002)	TC-Sec-02	6/6	Argon2 hasher; sensitive_post_parameters on auth views	**Passed**
TLS 1.2+ on every client-server connection (OEP-SR-001)	TC-Sec-01	17/17	Settings tests + check_tls.sh run (tls_deployment.md)	**Passed**
Server-side RBAC on every endpoint (OEP-F-003, OEP-SR-006)	TC-Auth-03, TC-Sec-06	26/26	test_rbac.py walks every /api/ route; endpoint_roles.md	**Passed**
Answer-key exposure checks (OEP-SR-004)	TC-Sec-04	15/15	Allow-listed student serializer; payloads checked for correct_answer	**Passed**
Account lockout / brute force after 5 failures (OEP-F-004, OEP-SR-005)	TC-Auth-04, TC-Sec-05	16/16	Lockout and audit-entry tests	**Passed**
Session expiry and token invalidation (OEP-SR-003)	TC-Sec-03	14/14	Idle-timeout and logout tests	**Passed**

§5.1 check	Test case	Automated passed	Evidence	Status
Fuzz testing of registration, login and question-import inputs	TC-Auth-01, TC-Auth-02, TC-QB-03	53/53	Boundary and malformed-input tests (bad JSON, extra columns, oversize files, invalid fields). No automated fuzzing tool was run.	**Partly passed**
Penetration testing of auth and exam-delivery endpoints (OWASP ZAP)	—	—	Not run: needs a deployed staging server. Covered in part by RBAC, answer-key and lockout tests.	**Not executed**

7. Exit criteria (Test Plan §5)

Criterion	Target	Actual	Met?
High-priority test cases executed	100%	23/25 (92.0%)	No (see §8)
Critical / high-severity defects open	0	0	Yes
Medium/Low-priority test cases executed	≥95%	18/18 (100.0%)	Yes
RTM shows every requirement traced to a test case	Yes	Yes, see §3 and the Test Case Document	Yes

8. Open items and deviations

These are the items not fully validated, with the reason. All are also recorded in `deviations.md` for the final demo.

Test case	Requirement	Status	Reason / next step
TC-Exam-05	OEP-F-024	Not executed	OEP-F-024 (course assignment, publish/unpublish) was moved to the backlog. Documented deviation; affects TC-Eval-02 and TC-Result-02 scoping.
TC-Perf-01	OEP-NF-001	Not executed	Needs a PostgreSQL database. Run exam_delivery/test_nfr.py against PostgreSQL and record P50/P90/P99 in docs/performance_report.md.
TC-UX-01	OEP-NF-005	Partly passed	Run the 3 moderated sessions in docs/usability_test.md and record the results.
TC-Arch-01	OEP-NF-006	Partly passed	Team signs off the architecture review; open gaps are SQLite (no shared DB for several instances) and missing service layers in three apps.

9. Conclusion

All automated tests pass and no defects are open. The core use cases (sign-up and login, question bank, exam assembly and scheduling, timed delivery with autosave and auto-submit, objective and subjective grading, scorecards and reports, admin management and diagnostics) are validated. The items in §8 need a production-like database (PostgreSQL), real participants or a team sign-off, or are deferred features; they are documented deviations, not failures.

10. Sign-off

Role	Name	Signature / Date
QA Lead	Rehan S Bawakhan	
Test Engineer	Yeshaswinie D	
Test Engineer	Yashwantha R	
Test Engineer	Yashaswini Singh	
Product Owner / Course Coordinator	Prof. Pavithra	